Mission Report - Team Rocket

Introduction

This report describes the digital signal processing (DSP) pipeline used to recover telemetry data from the Voyager-X probe's backup beacon signal. The Deep Space Network captured a weak radio signal embedded in noise. Our objective was to detect the signal, estimate the carrier frequency, correct the carrier offset, recover symbol timing, identify the modulation scheme, and extract the transmitted data.

1. Handling Memory Constraints (~15 GB Dataset)

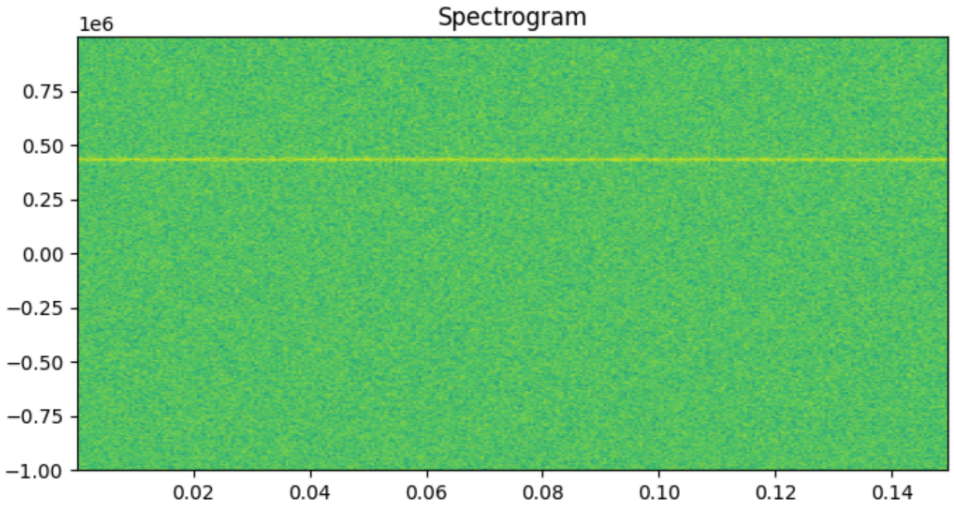
The dataset consisted of approximately 300 text files containing hexadecimal representations of IEEE-754 floating-point IQ samples. Since loading the entire dataset would exceed system memory limits, a streaming approach was implemented.

Key techniques used:

- Files processed sequentially using glob
- Each file read line-by-line
- Hex values converted to floats and combined as complex samples (I + jQ)
- Sample limits used during debugging to avoid RAM overflow
- Only necessary segments processed for spectral analysis

2. Spectrogram / Waterfall Analysis

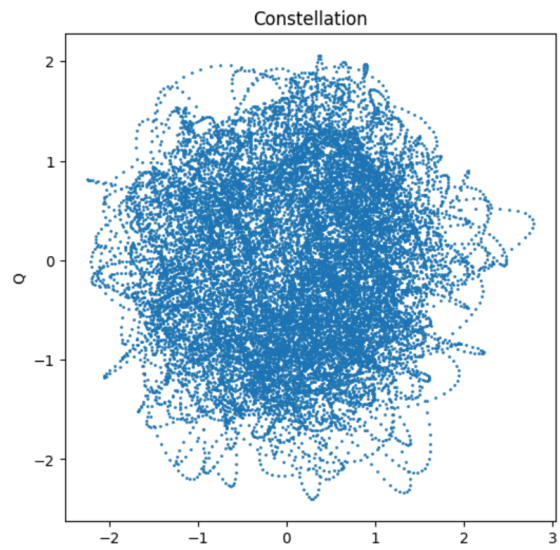
A spectrogram (short-time Fourier transform) was used to detect the presence of the signal and observe frequency behavior over time. A faint spectral line revealed the carrier around ~430 kHz above the noise floor.

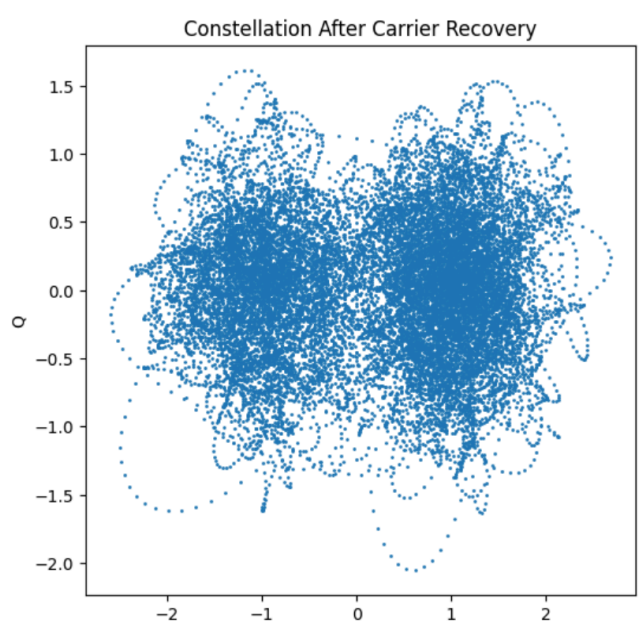


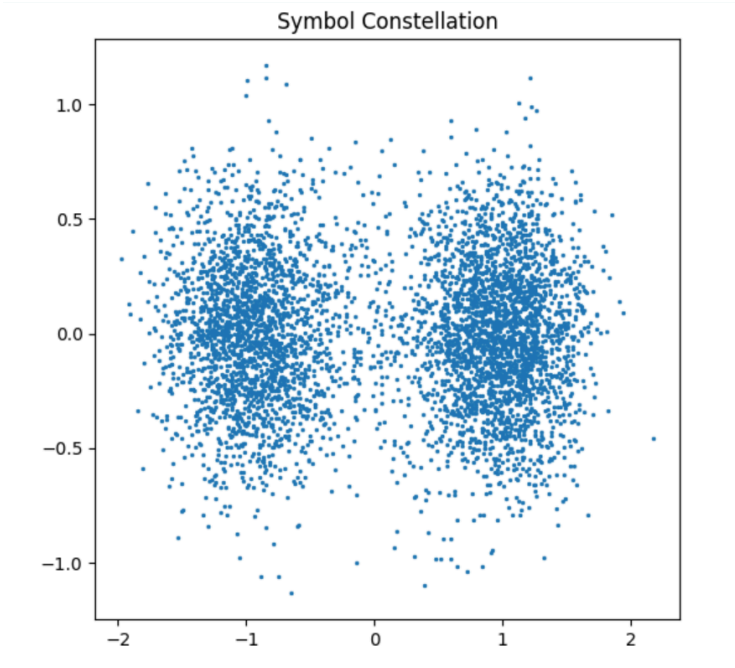
3. Carrier Recovery and Constellation Analysis

After detecting the carrier, the signal was shifted to baseband by multiplying the signal with a complex exponential. This removed the carrier frequency offset and allowed us to analyze the modulation constellation in the I-Q plane.

The initial constellation appeared as a noisy circular cloud due to phase drift. After carrier correction, two clear clusters became visible, indicating that the modulation scheme is Binary Phase Shift Keying (BPSK).

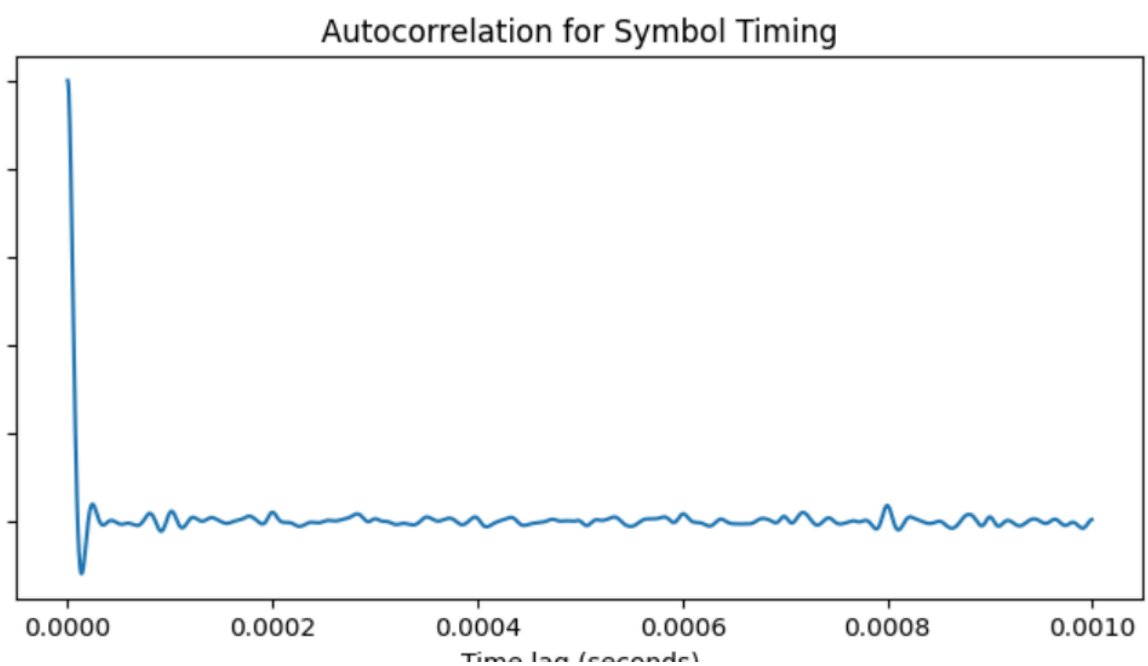




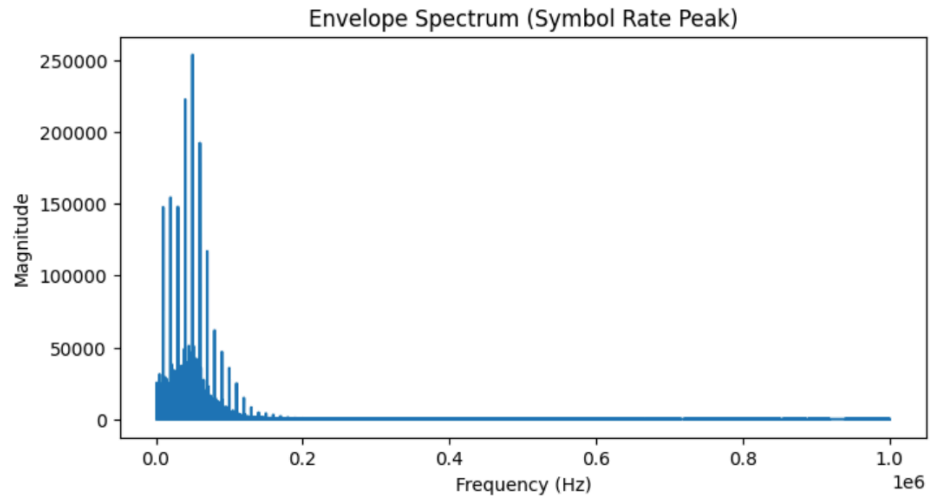


4. Symbol Timing Recovery

Autocorrelation and envelope spectrum analysis were used to estimate the symbol rate. The envelope spectrum revealed a clear peak corresponding to approximately 50,000 symbols per second. This allowed proper sampling of the received signal to recover symbol decisions.

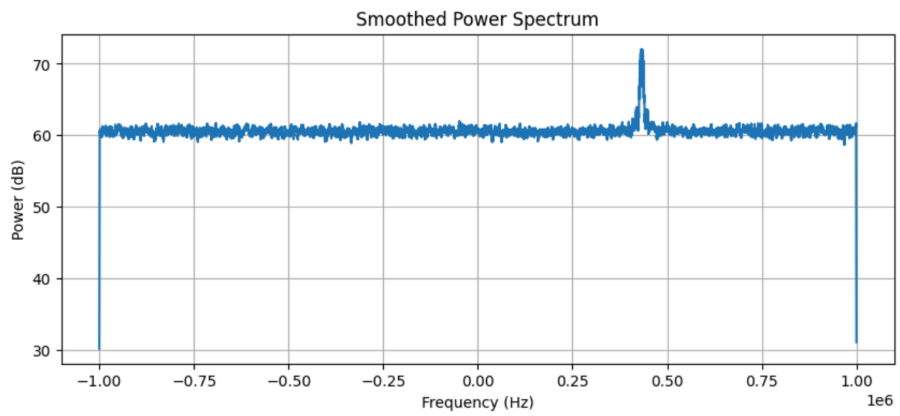


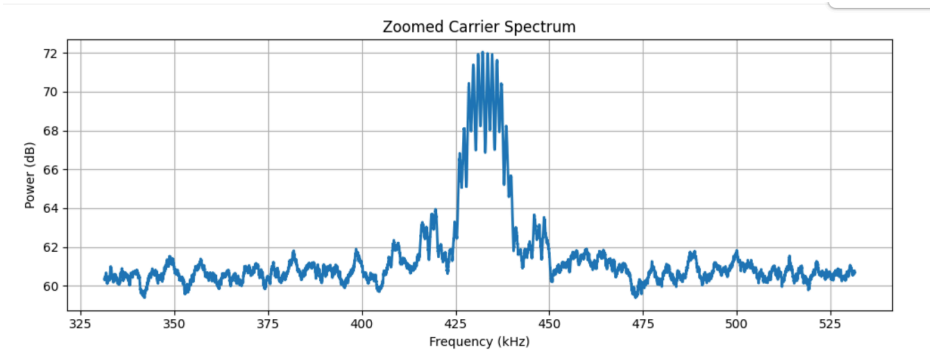
```
Estimated Symbol Rate: 50000.0
```

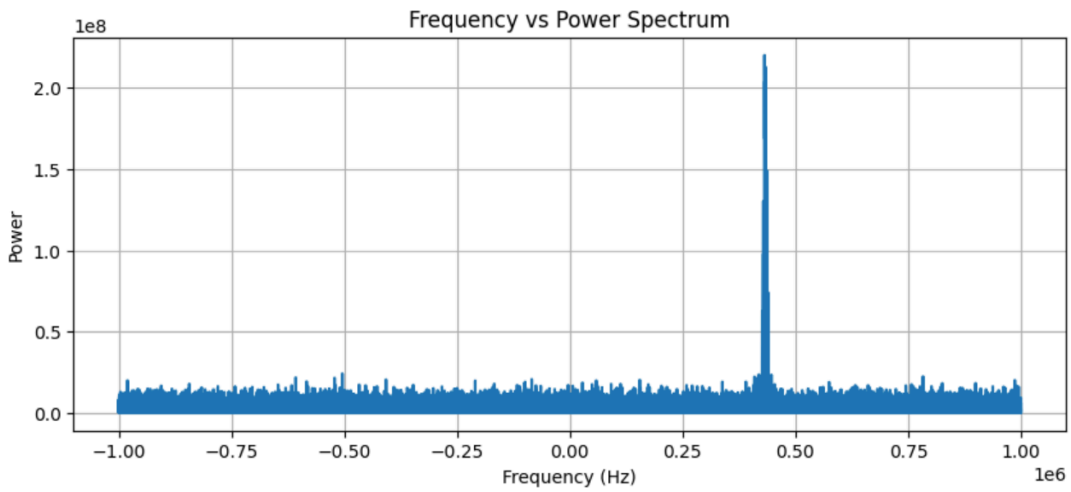


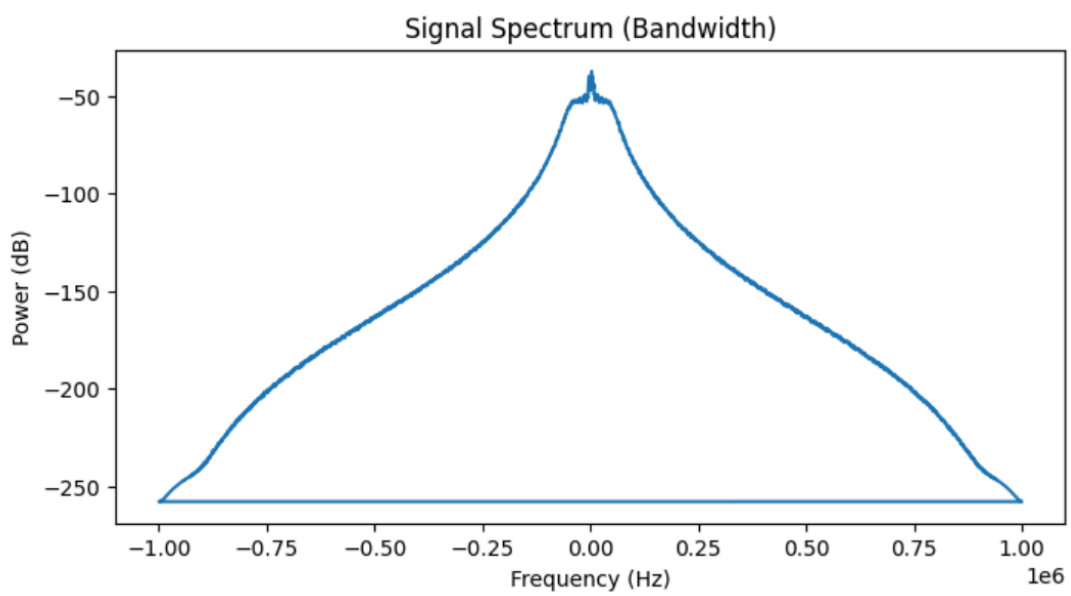
5. Data-Layer Processing

After symbol decisions were obtained, the resulting bitstream was analyzed for packet synchronization patterns. The CCSDS Attached Sync Marker (ASM) 0x1ACFFC1D was searched within the bitstream to identify frame boundaries. Once frames were aligned, payload data could be extracted for further analysis.









Conclusion

The implemented DSP pipeline successfully detected and recovered the Voyager-X beacon signal from a noisy dataset. Carrier detection, baseband conversion, symbol timing estimation, and constellation analysis confirmed a BPSK-modulated telemetry signal with an estimated symbol rate of approximately 50 kSymbols/s.